



Lab 13 - Open Ended Lab - Report

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Section: BS-AI

Course: Programming Fundamentals – Lab

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Task- 1

Output

Code – 1

0x94fc5ff6cc

0x94fc5ff6c8

0x94fc5ff6c4

Code – 2

5

Code – 3

5

0x8abe9ff984

0x8abe9ff984

5

Code – 4 (After declaring `int *ptr2`)

0x2938dffce0

0x2938dffce0

0x2938dffce4

0x2938dffce4

0x2938dffcdc

Code – 5

1

2

10

2

Task – 2

```
#include<iostream>
```

```
using namespace std;
```

```
void print ( int *ptr, int value )
```

```
{
```

```
    cout << "Value of the integer variable: " << value  
<< endl;
```

```
    cout << "Value obtained by dereferencing the  
pointer: " << *ptr << endl;
```

```
}
```

```
int main()
```

```
{
```

```
    int var = 10;
```

```
    int *p = &var;
```

```
    print ( p, var );
```

```
    return 0;
}
```

Output

```
PS E:\Lab Tasks\Open Ended Lab> ./13.2.exe
Value of the integer variable: 10
Value obtained by dereferencing the pointer: 10
```

Task – 3

```
#include<iostream>
using namespace std;
```

```
void print_reverse ( int *ptr, int size )
{
    ptr = ptr + ( size - 1 );
```

```
    for ( int ind = 0; ind < 5; ind++ )
```

```
{  
    cout << *ptr << endl;  
    ptr--;  
}  
}
```

```
int main()  
{  
    int arr [ 5 ] = { 10, 20, 30, 40, 50 };  
    int *p = arr;  
  
    print_reverse ( p, 5 );  
  
    return 0;  
}
```

Output

```
PS E:\Lab Tasks\Open Ended Lab> ./13.3.exe
50
40
30
20
10
```

Task – 4

```
#include<iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    float var = 3.9;
```

```
    float *p = &var;
```

```
    cout << "Original Value: " << var << endl;
```

```
    *p = 4.9;
```

```
    cout << "Updated Value: " << *p << endl;

    return 0;
}
```

Output

```
PS E:\Lab Tasks\Open Ended Lab> ./13.4.exe
Original Value: 3.9
Updated Value: 4.9
```

Task – 5

```
#include<iostream>

using namespace std;

void swap ( int *ptr1, int *ptr2 )
{
    int temp;
    temp = *ptr1;
```

```
*ptr1 = *ptr2;  
*ptr2 = temp;  
}
```

```
int main()  
{  
    int var1 = 10;  
    int var2 = 20;  
  
    cout << "---Before Swapping---" << endl;  
    cout << "Value 1: " << var1 << endl;  
    cout << "Value 2: " << var2 << endl;  
  
    swap ( &var1, &var2 );  
  
    cout << "---After Swapping---" << endl;  
    cout << "Value 1: " << var1 << endl;
```

```
    cout << "Value 2: " << var2 << endl;

    return 0;
}
```

Output

```
PS E:\Lab Tasks\Open Ended Lab> ./13.5.exe
---Before Swapping---
Value 1: 10
Value 2: 20
---After Swapping---
Value 1: 20
Value 2: 10
```

THE END